

Functional Dependencies:

- Airline
 - OP_CARRIER_AIRLINE_ID → OP_UNIQUE_CARRIER
 - From airline lookup file: OP_CARRIER_AIRLINE_ID → NAME
- Airport
 - AIRPORT_ID → IATA_CODE, CITY_NAME, STATE_ABR
 - From airport lookup: AIRPORT_ID → NAME
- Time Period
 - FL_DATE → YEAR, MONTH, DAY_OF_MONTH, DAY_OF_WEEK, QUARTER
- State
 - From state lookup file: STATE_ABR → STATE_NM
- Cancellation
 - From cancellation lookup file: CANCELLATION_CODE → DESCRIPTION
- Route
 - (ORIGIN_AIRPORT_ID, DEST_AIRPORT_ID) → DISTANCE
- Delay Type
 - DELAY_ID → NAME
- Diversion Details
 - For each diversion (1-5): (FLIGHT_ID, DIVERSION_NUMBER) → DIV_AIRPORT_ID, DIV_TOTALGTIME

First Normal Form (1NF):

A relation is in 1NF if all attributes contain atomic values and there are no repeating groups. Most attributes in the original dataset already contained atomic values. However, diversion information and delay information were identified as repeating groups.

Diversion Information:

The BTS dataset provides diversion information using repeated attributes:

- DIV1_AIRPORT_ID
- DIV1_TOTAL_GTIME
- DIV2_AIRPORT_ID
- DIV2_TOTAL_GTIME
- DIV3_AIRPORT_ID
- DIV3_TOTAL_GTIME
- DIV4_AIRPORT_ID
- DIV4_TOTAL_GTIME
- DIV5_AIRPORT_ID
- DIV5_TOTAL_GTIME

These attributes represent multiple occurrences of the same concept, namely a diverted airport. To eliminate the repeating group, diversion data was reorganized into:

FlightDiversion(

FLIGHT_ID FK,
DIVERSION_NUMBER,
DIV_AIRPORT_ID FK,

DIV_TOTAL_GTIME,
)
 Candidate and primary key: (FLIGHT_ID, DIVERSION_NUMBER)

For example:

Before:

FLIGHT_ID	DIV1_AIRPORT_ID	DIV2_AIRPORT_ID
1001	10004	10015

After:

FLIGHT_ID	DIVERSION_NUMBER	DIV_AIRPORT_ID
1001	1	10004
1001	2	10015

Delay Information:

The original dataset stores five delay categories as separate columns:

- CARRIER_DELAY
- WEATHER_DELAY
- NAS_DELAY
- SECURITY_DELAY
- LATE_AIRCRAFT_DELAY

These columns represent the same concept of delay minutes classified by delay type. To remove the repeating group, they were transformed into:

DelayType(
 DELAY_ID,
 NAME
)

For example:

Before:

FLIGHT_ID	CARRIER_DELAY	WEATHER_DELAY
1001	20	15

After:

FLIGHT_ID	DELAY_ID	MINUTES
1001	Carrier	20
1001	Weather	15

Now all attributes become atomic and the schema satisfies 1NF.

Second Normal Form (2NF):

A relation is in 2NF if it is already in 1NF and all non-key attributes are fully dependent on the whole primary key.

Several groups of attributes were dependent on entities other than the flight itself.

Airline dependencies: $OP_CARRIER_AIRLINE_ID \rightarrow OP_UNIQUE_CARRIER$

Therefore, we create:

```
Airlines(  
    OP_CARRIER_AIRLINE_ID PK,  
    OP_UNIQUE_CARRIER  
)
```

Airport dependencies: $AIRPORT_ID \rightarrow IATA_CODE, CITY_NAME, STATE_ABR$

Therefore, we create:

```
Airports(  
    AIRPORT_ID PK,  
    IATA_CODE,  
    CITY_NAME,  
    STATE_ABR  
)
```

Route dependencies: $(ORIGIN_AIRPORT_ID, DEST_AIRPORT_ID) \rightarrow DISTANCE$

The route distance depends on the airport pair from the origin to the destination, which is a candidate composite key. However, ROUTE_ID was introduced as a surrogate primary key to simplify foreign-key references from the Flights table.

Therefore, we create:

```
Route(  
    ROUTE_ID PK,  
    ORIGIN_AIRPORT_ID FK,  
    DEST_AIRPORT_ID FK,  
    DISTANCE  
)
```

Time Period dependencies: $FL_DATE \rightarrow YEAR, MONTH, DAY_OF_MONTH, DAY_OF_WEEK, QUARTER$

Therefore, we create:

```
TimePeriod(  
    FL_DATE PK,  
    YEAR,  
    MONTH,  
    DAY_OF_MONTH,  
    DAY_OF_WEEK,  
    QUARTER  
)
```

Flight Entity:

Once airline, route, and time period information were separated, the remaining identifying flight information became:

```
Flights(  
    Flight_ID PK,  
    OP_CARRIER_FL_NUM,  
    TAIL_NUM,  
    OP_CARRIER_AIRLINE_ID FK,  
    ROUTE_ID FK,  
    FL_DATE FK,  
    CANCELLED,  
    DIVERTED  
)
```

Flight Performance dependency:

The remaining operational attributes describe the performance of an individual flight occurrence. These attributes depend entirely on the flight itself and not on airlines, airports, routes, states, or dates.

The functional dependency here is:

- FLIGHT_ID → CRS_DEP_TIME, DEP_TIME, DEP_DELAY, DEP_DELAY_NEW, DEP_DEL15, TAXI_OUT, WHEELS_OFF, WHEELS_ON, TAXI_IN, CRS_ARR_TIME, ARR_TIME, ARR_DELAY, ARR_DELAY_NEW, ARR_DEL15, CRS_ELAPSED_TIME, ACTUAL_ELAPSED_TIME, AIR_TIME, CANCELLATION_CODE, DIV_AIRPORT_LANDINGS, DIV_REACHED_DEST, DIV_ACTUAL_ELAPSED_TIME, DIV_ARR_DELAY, DIV_DISTANCE

Therefore, these attributes are placed into a separate relational table:

```
FlightPerformance(  
    FLIGHT_ID PK/FK,  
    CRS_DEP_TIME,  
    DEP_TIME,  
    DEP_DELAY,  
    DEP_DELAY_NEW,  
    DEP_DEL15,  
    TAXI_OUT,  
    WHEELS_OFF,  
    WHEELS_ON,  
    TAXI_IN,  
    CRS_ARR_TIME,  
    ARR_TIME,  
    ARR_DELAY,  
    ARR_DELAY_NEW,  
    ARR_DEL15,  
    CRS_ELAPSED_TIME,
```

```

    ACTUAL_ELAPSED_TIME,
    AIR_TIME,
    CANCELLATION_CODE FK,
    DIV_AIRPORT_LANDINGS,
    DIV_REACHED_DEST,
    DIV_ACTUAL_ELAPSED_TIME,
    DIV_ARR_DELAY,
    DIV_DISTANCE
)

```

This table stores all the operational and performance metrics for a flight occurrence while preventing the Flights table from becoming excessively wide.

This removes partial dependencies and satisfies 2NF.

Third Normal Form (3NF):

A relation is in 3NF if there are no transitive dependencies.

Several transitive dependencies existed.

State dependency:

Airport records contained STATE_ABR, where $AIRPORT_ID \rightarrow STATE_ABR$, and since $STATE_ABR \rightarrow STATE_NM$, the state name was transitively dependent on the airport.

Therefore, we create:

```

StateNames(
    STATE_ABR PK,
    STATE_NM
)

```

Airline Name dependency:

Airline names are determined by airline identifiers, where $OP_CARRIER_AIRLINE_ID \rightarrow NAME$. So we create:

```

AirlineNames(
    OP_CARRIER_AIRLINE_ID PK/FK,
    NAME
)

```

Airport Name dependency:

Airport names are determined by airport identifiers, where $AIRPORT_ID \rightarrow NAME$. so we create:

```

AirportNames(
    AIRPORT_ID PK/FK,
    NAME
)

```

Cancellation dependency:

Cancellation descriptions depend on cancellation codes, where CANCELLATION_CODE → DESCRIPTION. So we create:

```
CancellationReason(  
    CANCELLATION_CODE PK,  
    DESCRIPTION  
)
```

Delay Type Dependency:

Delay category names depend on delay identifiers, where DELAY_ID → NAME. So we create:

```
DelayType(  
    DELAY_ID PK,  
    NAME  
)
```

The actual delay values are stored in:

```
FlightDelay(  
    FLIGHT_ID FK,  
    DELAY_ID FK,  
    MINUTES  
)
```

The candidate and primary key is (FLIGHT_ID, DELAY_ID)

This allows a flight to be associated with multiple delay categories.

After removing all transitive dependencies, the database satisfies the third normal form (3NF).

Final Schema:

```
TimePeriod(  
    FL_DATE PK,  
    YEAR,  
    MONTH,  
    DAY_OF_MONTH,  
    DAY_OF_WEEK,  
    QUARTER  
)
```

```
Airlines(  
    OP_CARRIER_AIRLINE_ID PK,  
    OP_UNIQUE_CARRIER  
)
```

```
AirlineNames(  
    OP_CARRIER_AIRLINE_ID PK/FK,  
    NAME  
)
```

```
StateNames(  
    STATE_ABR PK,  
    STATE_NM  
)
```

```
Airports(  
    AIRPORT_ID PK,  
    IATA_CODE,  
    CITY_NAME,  
    STATE_ABR FK  
)
```

```
AirportNames(  
    AIRPORT_ID PK/FK,  
    NAME  
)
```

```
Routes(  
    ROUTE_ID PK,  
    ORIGIN_AIRPORT_ID FK,  
    DEST_AIRPORT_ID FK,  
    DISTANCE  
)
```

```
Flights(  
    FLIGHT_ID PK,  
    OP_CARRIER_FL_NUM,  
    TAIL_NUM,  
    OP_CARRIER_AIRLINE_ID FK,  
    ROUTE_ID FK,  
    FL_DATE FK,  
    CANCELLED,  
    DIVERTED  
)
```

```
FlightPerformance(  
    FLIGHT_ID PK/FK,  
    CRS_DEP_TIME,  
    DEP_TIME,  
    DEP_DELAY,  
    DEP_DELAY_NEW,  
    DEP_DEL15,  
    TAXI_OUT,  
    WHEELS_OFF,
```

```

WHEELS_ON,
TAXI_IN,
CRS_ARR_TIME,
ARR_TIME,
ARR_DELAY,
ARR_DELAY_NEW,
ARR_DEL15,
CRS_ELAPSED_TIME,
ACTUAL_ELAPSED_TIME,
AIR_TIME,
CANCELLATION_CODE FK,
DIV_AIRPORT_LANDINGS,
DIV_REACHED_DEST,
DIV_ACTUAL_ELAPSED_TIME,
DIV_ARR_DELAY,
DIV_DISTANCE
)

CancellationReason(
    CANCELLATION_CODE PK,
    DESCRIPTION
)

DelayType(
    DELAY_ID PK,
    NAME
)

FlightDelay(
    FLIGHT_ID FK,
    DELAY_ID FK,
    MINUTES,
    PK(FLIGHT_ID, DELAY_ID)
)

FlightDiversion(
    FLIGHT_ID FK,
    DIVERSION_NUMBER,
    DIV_AIRPORT_ID FK,
    DIV_TOTAL_GTIME,
    PK(FLIGHT_ID, DIVERSION_NUMBER)
)

```

There are a total of 13 tables with 51 unique attributes.